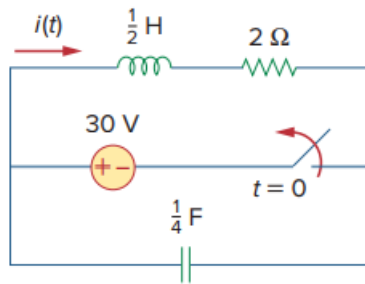


Problem

The switch in the circuit of Fig. 8.74 has been closed for a long time but is opened at $t = 0$. Determine $i(t)$ for $t > 0$.

Figure 8.74:



Step-by-step solution

Step 1 of 8

Refer to Figure 8.74 in the textbook.

At $t = 0^-$, switch is closed, the circuit reaches steady state. Inductor acts as short circuited and capacitor acts as open circuited.

Draw the modified circuit to calculate the initial values.

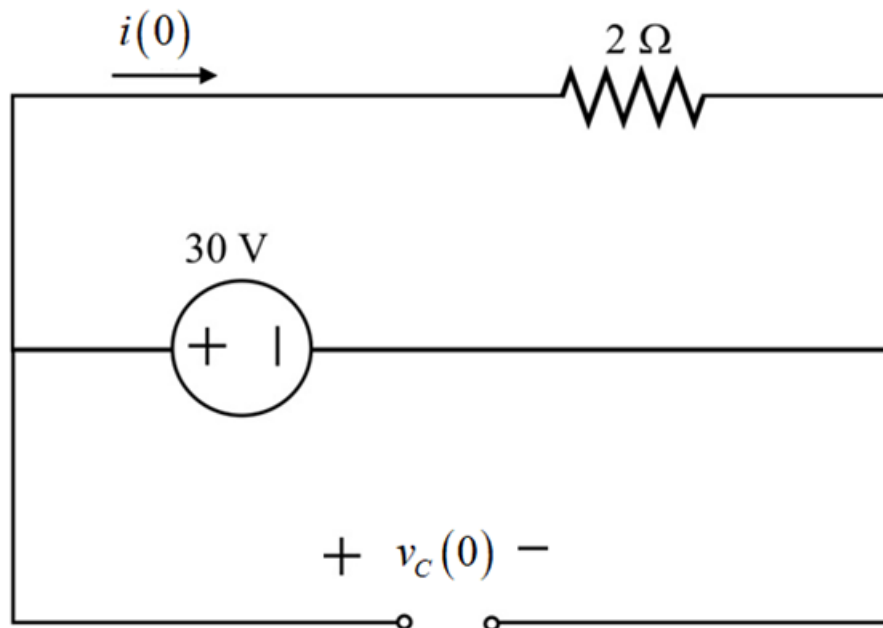


Figure 1

Step 2 of 8

The initial voltage across the capacitor is,

$$v_C(0) = 30 \text{ V}$$

Calculate the initial current through the inductor, $i(0)$.

$$\begin{aligned} i(0) &= \frac{30}{2} \\ &= 15 \text{ A} \end{aligned}$$

Step 3 of 8

Draw the circuit diagram for $t > 0$.

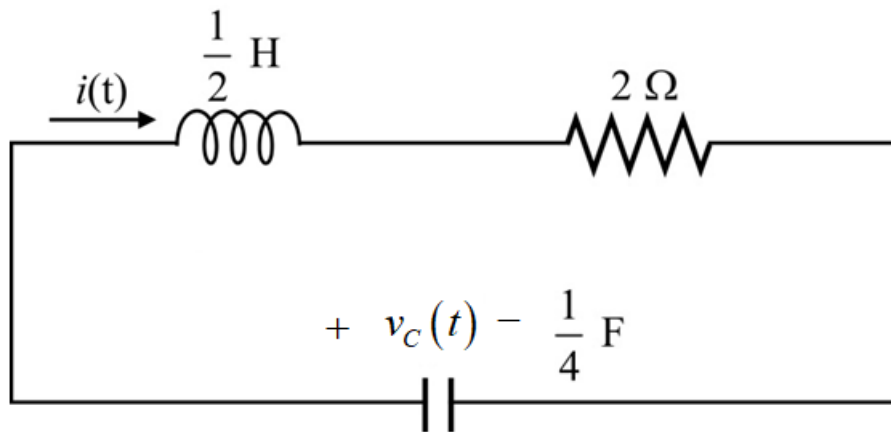


Figure 2

Step 4 of 8

From Figure 2, it is clear that the circuit is a RLC series circuit.

Calculate the value of Neper frequency.

$$\begin{aligned} \alpha &= \frac{R}{2L} \\ &= \frac{2}{(2)\left(\frac{1}{2}\right)} \\ &= 2 \end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)}} \\ &= \sqrt{8} \end{aligned}$$

Step 5 of 8

Here, $\omega_0 > \alpha$, so the response is under-damped.

Calculate the damped frequency.

$$\begin{aligned}\omega_d &= \sqrt{\omega_0^2 - \alpha^2} \\ &= \sqrt{(2.828)^2 - 2^2} \\ &= \sqrt{8 - 4} \\ &= 2\end{aligned}$$

Consider the expression for the inductor current.

$$i(t) = I_s + e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t) \text{ A}$$

For $t > 0$, the circuit is a source-free circuit.

$$I_s = 0 \text{ A}$$

Substitute 0 A for I_s , 2 for α and 2 for ω_d in $i(t)$.

$$i(t) = 0 + e^{-2t} (A_1 \cos 2t + A_2 \sin 2t) \text{ A}$$

$$i(t) = e^{-2t} (A_1 \cos 2t + A_2 \sin 2t) \text{ A} \dots\dots (1)$$

Step 6 of 8

For $t = 0$,

$$\begin{aligned}i(0) &= e^{-2(0)} (A_1 \cos 2(0) + A_2 \sin 2(0)) \\ &= A_1\end{aligned}$$

Substitute 15 for $i(0)$ in the equation.

$$15 = A_1$$

$$A_1 = 15$$

Step 7 of 8

Write the Kirchhoff's voltage to the loop.

$$L \frac{di(t)}{dt} + Ri(t) = v_c(t)$$

$$\frac{di(t)}{dt} = \frac{v_c(t)}{L} - \frac{R}{L} i(t)$$

At $t = 0$,

$$\begin{aligned}\frac{di(0)}{dt} &= \frac{v_c(0)}{L} - \frac{R}{L} i(0) \\ &= \frac{30}{\left(\frac{1}{2}\right)} - \frac{2}{\left(\frac{1}{2}\right)} (15) \\ &= 60 - 60 \\ &= 0\end{aligned}$$

Step 8 of 8

Differentiate equation (1), $i(t) = e^{-2t} (A_1 \cos 2t + A_2 \sin 2t)$

$$\begin{aligned}\frac{di(t)}{dt} &= e^{-2t} (-2A_1 \sin 2t + 2A_2 \cos 2t) - 2e^{-2t} (A_1 \cos 2t + A_2 \sin 2t) \\ &= e^{-2t} (-2A_1 \sin 2t + 2A_2 \cos 2t - 2A_1 \cos 2t - 2A_2 \sin 2t)\end{aligned}$$

At $t = 0$,

$$\frac{di(0)}{dt} = e^{-2(0)} (-2A_1 \sin 2(0) + 2A_2 \cos 2(0) - 2A_1 \cos 2(0) - 2A_2 \sin 2(0))$$

$$0 = 2A_2 - 2A_1$$

$$A_2 = A_1$$

$$A_2 = 15$$

Substitute 15 for A_1 and 15 for A_2 in $i(t) = e^{-2t} (A_1 \cos 2t + A_2 \sin 2t)$.

$$\begin{aligned}i(t) &= e^{-2t} (15 \cos 2t + 15 \sin 2t) \\ &= 15e^{-2t} (\cos 2t + \sin 2t) \text{ A}\end{aligned}$$

Thus, the current $i(t)$ for $t > 0$ is $\boxed{15e^{-2t} (\cos 2t + \sin 2t) \text{ A}}$.